



Gastric Mucosa: Regions & Cells

Histology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 9 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.**EASY**

Into how many regions is the stomach divided in this lecture?

- A. Three regions
- B. Five regions
- C. Four regions
- D. Two regions
- E. Six regions

2.**EASY**

Which gastric cell secretes hydrochloric acid?

- A. Stem cell
- B. Parietal cell
- C. Mucous neck cell
- D. Surface mucous cell
- E. Chief cell

3.**EASY**

Which gastric cell secretes pepsinogen?

- A. Stem cell
- B. Surface mucous cell
- C. Chief cell
- D. Parietal cell
- E. Mucous neck cell

4.**EASY**

Rugae are:

- A. mucosa/submucosa folds that flatten when the stomach fills
- B. finger-like villi of the gastric mucosa
- C. folds of the muscularis externa in the pylorus
- D. epithelial invaginations that hold the glands
- E. the thickened pyloric sphincter ring

5.**MEDIUM****CASE**

A gastric section shows shallow pits opening from long, branched tubular glands whose lower halves contain many basophilic cells and some pale round cells.

Which region is this?

- A. Esophagus
- B. Duodenum
- C. Pyloric region
- D. Fundus or body
- E. Cardia region

6.**MEDIUM**

Which parietal-cell product is required for vitamin B12 absorption?

- A. Mucin
- B. Pepsinogen
- C. Intrinsic factor
- D. Gastrin
- E. Gastric lipase

7.

MEDIUM

CASE

At the base of a gastric gland lies a basophilic cell packed with rough endoplasmic reticulum and membrane-limited granules of an inactive enzyme.

Which cell is this?

- A. Mucous neck cell
- B. Stem cell
- C. Parietal cell
- D. Chief cell
- E. Enteroendocrine cell

8.

MEDIUM

Which feature helps protect the gastric epithelium from acid?

- A. A keratinized layer covering the surface
- B. A thick mucus layer from all surface and pit cells, plus tight junctions
- C. A stratified squamous surface epithelium
- D. A brush border of digestive enzymes
- E. A mesothelial covering over the mucosa

9.

MEDIUM

CASE

In the neck of a gastric gland, low columnar cells with oval nuclei near their base show a high rate of mitosis and increase in number next to an ulcer.

Which cell is this?

- A. Stem cell
- B. Chief cell
- C. Mucous neck cell
- D. Parietal cell
- E. Enteroendocrine cell

10.

MEDIUM

Which distribution of cells in a fundic gland is correct?

- A. Neck: stem, mucous neck, parietal; base: chief, enteroendocrine, few parietal
- B. Neck: chief cells only; base: parietal cells only
- C. Neck: enteroendocrine cells only; base: surface mucous cells
- D. Uniform: every cell type spread evenly along the gland
- E. Neck: chief and enteroendocrine; base: stem and mucous neck cells

11.

MEDIUM

CASE

A pyloric biopsy contains mostly mucus-secreting cells, scarce chief cells, and scattered enteroendocrine cells.

Which cells release gastrin and which release somatostatin?

- A. Both are released by mucous neck cells
- B. D cells release gastrin; G cells release somatostatin
- C. Chief cells release gastrin; parietal cells release somatostatin
- D. K cells release gastrin; S cells release somatostatin
- E. G cells release gastrin; D cells release somatostatin

12.

MEDIUM

The muscularis externa of the stomach consists of:

- A. two layers only: inner circular and outer longitudinal
- B. three smooth layers: outer longitudinal, middle circular, inner oblique
- C. three layers of skeletal muscle in the wall
- D. an outer circular and an inner longitudinal layer
- E. a single circular layer around the lumen

13.

MEDIUM

CASE

Near the opening of the esophagus, a small area of mucosa has shallow pits and branched tubular glands whose secretory cells produce mucus; a few parietal cells are present.

Which region is this?

- A. Duodenum
- B. Pylorus
- C. Body
- D. Fundus
- E. Cardia

14.

MEDIUM

The lamina propria of the stomach contains:

- A. keratinized epithelium over dense CT
- B. skeletal muscle covered by mesothelium
- C. loose CT with smooth muscle, lymphoid cells and glands
- D. blood vessels only, with no glands
- E. dense CT with Meissner's plexus and large vessels

15.

HARD

CASE

Section A: shallow pits with long glands containing parietal and chief cells. Section B: long pits, short glands, predominantly mucus-secreting cells with scarce chief cells.

Which pairing is correct?

- A. A = pylorus; B = cardia
- B. A = fundus / body; B = pylorus
- C. A = cardia; B = fundus / body
- D. A = fundus / body; B = cardia
- E. A = pylorus; B = fundus / body

16.

HARD

CASE

A patient has lost all the parietal cells of the stomach.

Which two consequences are expected?

- A. Reduced conversion of pepsinogen to pepsin, and impaired vitamin B12 absorption
- B. Failure to secrete pepsinogen, and loss of gastric lipase
- C. Failure of epithelial renewal, and loss of gastric lipase
- D. Loss of the protective mucus layer, and impaired B12 absorption
- E. Loss of gastrin, and failure of the pyloric sphincter to open

17.

HARD

CASE

After a gastric ulcer the surface epithelium regenerates.

Which cell population heals the defect and what is its turnover time in this lecture?

- A. Enteroendocrine cells at the base; about 3–4 days
- B. Chief cells at the base; about 4–7 days
- C. Stem cells in the neck; about 2–4 days
- D. Stem cells in the neck region; about 4–7 days
- E. Parietal cells in the neck; about 4–7 days

18.

HARD

CASE

Cell A is large, pale and round with 1–2 central nuclei and acidophilic cytoplasm packed with mitochondria. Cell B is smaller with basophilic cytoplasm rich in rough endoplasmic reticulum.

Which identification is correct?

- A. A = mucous neck cell; B = chief cell
- B. A = chief cell (pepsinogen); B = parietal cell (HCl, intrinsic factor)
- C. A = parietal cell; B = enteroendocrine cell
- D. A = surface mucous cell; B = parietal cell
- E. A = parietal cell (HCl, intrinsic factor); B = chief cell (pepsinogen)

19.

HARD

Which trio is correctly matched with its site and product?

- A. K cells — pylorus — gastrin; S cells — ileum — GIP; G cells — small intestine — secretin
- B. K cells — ileum — motilin; S cells — pylorus — secretin; G cells — colon — gastrin
- C. K cells — colon — peptide YY; S cells — pylorus — gastrin; G cells — jejunum — secretin
- D. K cells — duodenum and jejunum — GIP; S cells — small intestine — secretin; G cells — pylorus — gastrin
- E. K cells — small intestine — secretin; S cells — duodenum — GIP; G cells — ileum — gastrin

20.

HARD

Which two APUD cell types increase gut motility according to the lecture table?

- A. L cells and N cells
- B. I cells and K cells
- C. S cells and D cells
- D. G cells and D cells
- E. EC cells and Mo cells

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	C	Cardia, fundus, body and pyloric region.
2	B	Parietal cells produce HCl (and intrinsic factor).
3	C	Chief cells hold membrane-limited pepsinogen granules.
4	A	Rugae are mucosa + submucosa folds; they flatten when the stomach is filled with food.
5	D	Shallow pits with long branched tubular gastric glands and all the cell types = fundus / body (identical histologically).
6	C	Intrinsic factor is a glycoprotein required for B12 absorption.
7	D	Basophilic, RER-rich, base of the gland, pepsinogen granules = chief (zymogenic) cell.
8	B	Simple columnar surface and pit cells all secrete mucus forming a thick protective layer; tight junctions form part of the barrier.
9	A	Low columnar cells in the neck with a high mitotic rate that increase after damage are the stem cells.
10	A	The distribution of epithelial cells in a gastric gland is not uniform: stem / mucous neck / parietal in the neck; chief / enteroendocrine / few parietal at the base.
11	E	In the pylorus, G cells release gastrin and D cells secrete somatostatin.
12	B	Smooth muscle fibres run in three directions — outer longitudinal, middle circular, inner oblique.
13	E	A small area near the esophageal opening with shallow pits and mucus-secreting cardiac glands (few parietal cells) is the cardiac region.
14	C	The lamina propria is loose CT containing smooth muscle, lymphoid cells and the glands.
15	B	Shallow pits + long glands = fundus / body; long pits + short mucus-cell glands = pylorus.
16	A	Parietal cells make HCl (needed to convert pepsinogen to pepsin) and intrinsic factor (needed for B12 absorption).
17	D	Gastric stem cells (neck) have a turnover of 4-7 days and increase in number after damage such as an ulcer.
18	E	Large pale cells with central nuclei, acidophilic cytoplasm and many mitochondria = parietal; smaller basophilic RER-rich cells = chief.
19	D	From the lecture table: K = duodenum + jejunum (GIP); S = small intestine (secretin); G = pylorus (gastrin).
20	E	The table lists EC cells (serotonin and substance P) and Mo cells (motilin) with increased gut motility.

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Written from the lecture slides only – check any answer you doubt against your lecture notes.